

NeuroCore – Pipeline Guidelines (Controlled Document)

1. Scope

This document defines the structured guidelines for implementing a transition-sensitive pipeline based on the NeuroCore framework. It is intended for controlled technical evaluation and does not include full reproducibility parameters.

2. Pipeline Overview

The pipeline follows a multi-layer transformation from raw signal to regime inference:

$$x(t) \rightarrow \Phi(t) \rightarrow \Delta(t), K(t), L(t) \rightarrow z(t) \rightarrow \lambda(t)$$

3. Signal Input

Input consists of time-series data (e.g., EEG, fNIRS). Signals must be normalized and validated before processing.

```
# Load and normalize
X = load_signal()
X = normalize(X)
```

4. Local Operator (Φ)

Computes local gradient-based signal dynamics.

```
grad = abs(gradient(X))
phi = log(1 + grad)
# [REDACTED: window size and smoothing]
```

5. Transition Metric (Δ)

Measures temporal instability.

```
delta = abs(phi[t] - phi[t-1])
# [REDACTED: threshold logic]
```

6. Global Stability (K)

Represents dispersion across signal.

```
K = std(X) / mean(X)
# [REDACTED: stability constraints]
```

7. Asymptotic Stability (L)

Tracks convergence toward functional anchor (~0.55).

```
L = compute_L(X)
# target ≈ 0.55
# [REDACTED: convergence model]
```

8. Feature Fusion

Combines metrics into unified representation.

```
z = fuse(delta, K, L)
# [REDACTED: weights and logic]
```

9. Regime Inference

Final classification or segmentation step.

```
lambda_t = infer_regime(z)
# [REDACTED: classifier / rules]
```

10. Validation Guidelines

Validation must include:

- Proper train/test separation
- Event alignment verification
- Artefact rejection
- Avoidance of label leakage

11. Restrictions

Critical parameters, dataset handling, and full pipeline implementation are intentionally omitted.

12. Conclusion

This document provides a structured view of the NeuroCore pipeline while preserving the integrity of the protected operational core.